

ENHANCING RESTAURANT-LEVEL PROCESS CONSISTENCY & CUSTOMER EXPERIENCE

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01 | THE QUALITY CHALLENGE

McDonald's operates at an enormous global scale, where customer expectation demands not simply good food, but a consistent, dependable restaurant experience.

Key drivers include speed, accuracy, friendliness, and quality. However, restaurant-level experience still varies due to:

- Service and waiting time fluctuations
- Order accuracy inconsistencies
- Peak-hour operational bottlenecks
- Process gaps between standard & ground execution

PROBLEM STATEMENT

How can McDonald's strengthen restaurant-level process consistency, reduce avoidable service variation, and deliver a more reliable customer experience across locations?

02 | QUALITY ENGINEERING APPROACH

Prevention & Understanding First: A process cannot be continuously improved unless the organization understands what is actually going wrong at ground level.

Step 1: Voice of Customer (VOC) + System & Process Audit

- **VOC Collection:** Gather structured feedback to capture customer dissatisfaction, expectations, and recurring pain points. Translate actionable requirements into process improvements.
- **System & Process Audits:** Evaluate process adherence, service execution, order accuracy, food quality controls, and hygiene practices.
- **Audit Objective:** Identify where actual execution deviates from standard operating procedures (SOPs) to drive prevention.

03 | FROM PROBLEMS TO REAL ROOT CAUSE

When audits and VOC reveal multiple operational issues, the goal is structured prioritization rather than solving everything simultaneously.

1. Fishbone (Ishikawa) Analysis

Categorizes potential causes behind service variation across 6 key operational dimensions:

Man	Crew training, fatigue, communication
Machine	POS responsiveness, kitchen equipment layout
Method	Order assembly workflow, station handoffs
Material	Packaging, ingredient staging, inventory flow
Measurement	Order accuracy tracking, hold-time monitors
Environment	Peak-hour kitchen heat, aisle noise, space layout

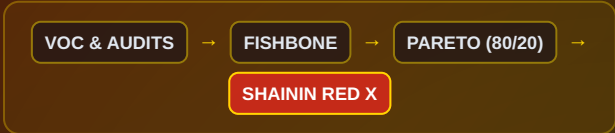
2. Pareto Analysis (80/20 Principle)

Applied as a prioritization concept to separate the "vital few" defects from the "trivial many", focusing engineering effort where impact is highest.

3. Shainin Red X Methodology

Applied post-prioritization to isolate the single dominant root cause (the **Red X**) responsible for critical variation, enabling targeted intervention.

VISUAL QE IDENTIFICATION FLOW



04 | SOLUTION DESIGN & INNOVATION

Finding the root cause is only half the problem. The process must be redesigned without shifting constraints elsewhere.

Theory of Constraints (TOC)

Identifies the critical bottleneck limiting peak-hour throughput. Improving non-bottleneck activities does not improve total customer experience; the primary constraint must be measured and optimized first.

TRIZ (Theory of Inventive Problem Solving)

Resolves core operational contradictions without compromise (e.g., *Increasing speed without reducing order accuracy*). Eliminates trade-off thinking.

05 | VALIDATION BEFORE IMPLEMENTATION

VALIDATE → PILOT → MEASURE → IMPROVE → IMPLEMENT

Controlled pilots are conducted in selected restaurants to verify performance under real operational conditions before full rollout.

PRIMARY PILOT TARGET METRIC

≥ 98%

COMPLETE ORDER ACCURACY RATE

Digitally tracked via POS integration and audit sampling

06 | FAILURE MODE ANALYSIS (FMEA)

New failure modes may surface during pilots. FMEA evaluates risks using Severity, Occurrence, and Detection.

- **Critical Severity Focus:** Identified customer experience failure mode evaluated at **Severity = 8/10**.
- **RPN Action:** High-impact risks are mitigated proactively prior to system-wide deployment.

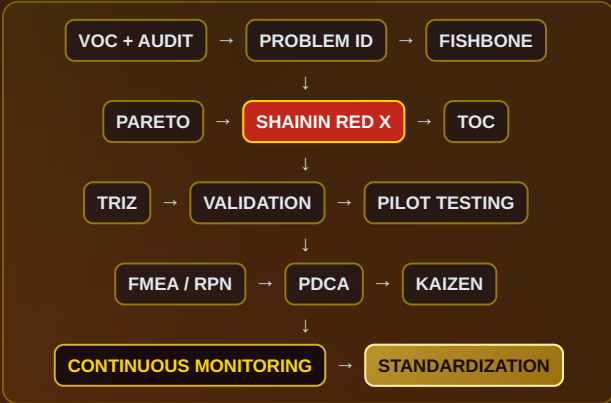
07 | CONTROL & CONTINUOUS IMPROVEMENT

PDCA + Kaizen + Continuous Monitoring

- **Plan:** Define problem, root cause, and ≥98% accuracy target.
- **Do:** Execute controlled pilot in selected locations.
- **Check:** Analyze POS data, digital tracking, and audit metrics.
- **Act:** Standardize successful SOPs; correct deviations.

Kaizen ensures ongoing incremental gains, while digital POS monitoring alerts management to drift.

10 | COMPLETE QE METHODOLOGY FLOW



08 | EXPECTED BUSINESS IMPACT

Customer	Operational	Business
• Consistent experience • Higher accuracy • Reduced wait times	• Controlled variation • Early failure detection • Faster bottleneck response	• Stronger brand trust • Scalable execution • Efficiency alignment

09 | QE CONCEPTS APPLIED



11 | CONCLUSION

The goal is not to create another SOP manual, but to make quality visible in actual restaurant execution. Customer feedback shows experience; audits reveal deviation; Fishbone & Pareto structure problems; Shainin isolates critical causes; TOC targets bottlenecks; TRIZ resolves trade-offs; FMEA anticipates risk; PDCA & Kaizen sustain performance. **That is execution-driven Quality Engineering.**

12 | Disclaimer

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